

### ✅ SQL Server Licensing Models – Simplified Table

| Edition              | Licensing Model                   | Best For                             | Pricing Basis                | Key Features / Notes                                           |
|----------------------|-----------------------------------|--------------------------------------|------------------------------|----------------------------------------------------------------|
| SQL Server Express   | Free                              | Students, Developers, Small Apps     | Free                         | Limited to 10 GB per DB, 1 GB RAM, 1 Socket or 4 Cores         |
| SQL Server Developer | Free (Non-production only)        | Learning, Dev & Testing Only         | Free                         | Same features as Enterprise, but <b>not for production use</b> |
| Standard Edition     | Core-based / Server + CAL         | Small to Medium Applications         | Per Core or Per Server + CAL | Up to 24 cores, high availability, basic BI                    |
| Enterprise Edition   | Core-based only                   | Large, Mission-critical Applications | Per Core only                | Full features: Always On, Advanced Security, BI, Big Data      |
| Web Edition          | Core-based (via hosting partners) | Web hosting providers                | Special pricing via SPLA     | Affordable for web hosting; only via service providers         |
| Azure SQL (PaaS)     | vCore or DTU-based                | Cloud-based SQL DB                   | Pay-as-you-go                | Fully managed, backups, HA, scaling in cloud                   |

### ✅ Licensing Models Explained:

| Model                                     | Description                                                                |
|-------------------------------------------|----------------------------------------------------------------------------|
| Per Core                                  | Pay per physical or virtual core (minimum 4 cores per server).             |
| Server + CAL                              | Buy a server license + Client Access Licenses (CALs) for each user/device. |
| SPLA (Service Provider License Agreement) | Monthly rental model for service providers.                                |
| Azure vCore/DTU                           | For Azure SQL — Pay by compute size (vCore) or bundled unit (DTU).         |

 **Real-Life Examples:**

| Scenario                       | Recommended License                    |
|--------------------------------|----------------------------------------|
| Learning SQL Server at home    | Developer Edition (Free)               |
| Small office with 10 employees | Standard Edition + 10 CALs             |
| Large banking application      | Enterprise Edition – Per Core          |
| Hosting SQL Database on Azure  | Azure SQL – vCore or DTU-based pricing |

## ✅ SQL Server Licensing: Per Core vs Server + CAL

| Feature                          | Per Core Licensing                                                                                                 | Server + CAL Licensing                                                                                                         |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Usage Type                       | Best for <b>highly concurrent users or external access</b>                                                         | Best for <b>limited, known internal users</b>                                                                                  |
| Applies To                       | SQL Server <b>Enterprise &amp; Standard Editions</b>                                                               | SQL Server <b>Standard Edition only</b>                                                                                        |
| Pricing Basis                    | Pay per <b>physical/virtual core</b> (minimum 4 core licenses)                                                     | Pay per <b>server license + CALs</b> for each user/device                                                                      |
| Core Requirements                | Minimum <b>4 core licenses per server</b>                                                                          | Only <b>1 server license + CALs</b>                                                                                            |
| Client Access Licenses (CALs)    | ❌ Not required                                                                                                     | ✅ <b>Required</b> for every user/device accessing SQL Server                                                                   |
| External Users / Internet Access | ✅ Allowed without extra cost                                                                                       | ❌ Each external user must have a CAL (not practical)                                                                           |
| Virtualization                   | Fully supported, especially in <b>Enterprise Edition</b>                                                           | Limited flexibility in virtual environments                                                                                    |
| Scalability                      | Highly scalable – pay as per CPU power                                                                             | Less scalable – CAL costs grow with number of users/devices                                                                    |
| Best Use Cases                   | <ul style="list-style-type: none"><li>- Public-facing apps</li><li>- Cloud</li><li>- Large organizations</li></ul> | <ul style="list-style-type: none"><li>- Small/medium businesses</li><li>- Internal use</li><li>- Fixed users/devices</li></ul> |
| License Cost Control             | Cost increases with CPU cores                                                                                      | Cost increases with number of users/devices                                                                                    |
| Cloud Usage (Azure, AWS)         | Preferred in cloud (e.g., per vCore)                                                                               | Not suitable for cloud environments                                                                                            |
| Example Scenario                 | E-commerce website with 5000+ users                                                                                | Internal ERP system with 10 employees using the database                                                                       |

## Summary

| Choose This If...                                              | Recommended Model |
|----------------------------------------------------------------|-------------------|
| You don't know the number of users or it's very high           | Per Core          |
| The SQL Server is used by external/public internet users       | Per Core          |
| You have a small team with fixed, countable internal users     | Server + CAL      |
| You need flexibility in virtualization or hybrid/cloud hosting | Per Core          |
| You want to minimize license cost in a small on-premise setup  | Server + CAL      |

Per Core and Server+CAL pricing in detail in SQL Server Licensing

Here is a detailed pricing comparison of Per Core vs Server + CAL licensing for SQL Server 2022 (Standard & Enterprise Editions) in a clear table format, using USD pricing (approximate retail prices) for understanding. (INR pricing available on request).

💰 SQL Server Licensing Pricing: Per Core vs Server + CAL

| Edition            | Licensing Model | Component                 | Approx. Price (USD)   | Notes                                                         |
|--------------------|-----------------|---------------------------|-----------------------|---------------------------------------------------------------|
| Standard Edition   | Per Core        | Per Core License (2-pack) | \$3,586               | Must buy <b>minimum 4 cores</b> (i.e., 2 × 2-pack) = ~\$7,172 |
|                    | Server + CAL    | Server License            | \$931                 | One license per server                                        |
|                    |                 | User CAL                  | \$209 per user/device | Required for <b>each</b> user/device accessing the SQL Server |
| Enterprise Edition | Per Core Only   | Per Core License (2-pack) | \$13,748              | Must buy minimum 4 cores (2 × 2-pack) = ~\$27,496             |
|                    | Server + CAL    | ✗ Not Available           | —                     | Enterprise only supports <b>Per Core model</b>                |

✅ Cost Examples

| Scenario                                      | Model                 | Components Needed                 | Total Cost (Approx.) |
|-----------------------------------------------|-----------------------|-----------------------------------|----------------------|
| Small team with 1 server & 5 users            | Server + CAL          | \$931 (server) + 5 × \$209 (CALs) | \$1,976              |
| Medium app with 1 server & 20 users           | Server + CAL          | \$931 + 20 × \$209                | \$5,111              |
| Heavy app with 1 server (8 cores), many users | Per Core (Standard)   | 4 × \$3,586 (i.e. 8 cores)        | \$14,344             |
| Mission-critical app (8 cores)                | Per Core (Enterprise) | 4 × \$13,748 (i.e. 8 cores)       | \$54,992             |



### Additional Notes:

- CALs are **not needed** in **Per Core** model.
- Each CAL is **user-based or device-based**, not concurrent.
- **Core licenses must be bought in packs of 2**, minimum 4 cores per server.
- Licensing costs can vary based on **resellers, region, and agreements**.